

Project (ENSE 375)

1. **Project Description:** In groups/teams of 3 (choose your own team members), teams will design, develop and comprehensively test a software application of their choice by using test-driven development methodology, Model-View Controller (MVC) architecture for code management and Junit for test cases (where applicable). Test your application using following techniques. (Remember, the project's main objective is to design the optimal test suit for the application).
 - Choose any function from the application and test it with
 - Path testing
 - Data flow
 - Choose a subset of units to perform Integration testing.
 - Perform the validation of your application by using the following:
 - Boundary value testing
 - Equivalence class testing
 - Decision tables testing
 - State transition testing
 - Use case testing.

Note: As mentioned, your team's proposed software application must be approved by the instructor and be of adequate scope for an approximate 2-month design/development time-box. Think small-to-medium project scope. Examples could include a howitzer firing simulator, text file encryption/decryption tool, file-based library management system, student grade analyzer, personal expense tracker, and task scheduler etc.

Hint: Follow along with what Lab Instructor has in store for you. The labs will greatly assist you while developing the application and writing the test cases. Best to follow along with the labs and learn!

2. **Design and Architecture:** You need to go through the structured engineering design process (Figure 1) while considering different factors that can impact the testing of the final product. Teams are encouraged to use the V-shape software development lifecycle.
3. **Design Constraints:** Your solution should consider if not all but at least four from the following list of design constraints
 - a. Economic factors
 - b. Regulatory compliance (Security and Access)
 - c. Reliability
 - d. Sustainability and Environmental Factors
 - e. Ethics
 - f. Societal Impacts
4. **Teamwork:** Each group must uphold the best teamwork strategy, including team formation, time management, and conflict resolution.
5. **Presentation Requirements:** A separate written report is **NOT** required, but the entire GitHub project should be 'self-documenting'
 - The project should include a **REPORT.md** file describing the use instructions of your application and providing other details according to the project file template (Project_File_Template.pdf) available on UR courses.

- The project should include a TESTING.md that describes the test plan. It should include any necessary information to understand the test cases. For example, in equivalence class testing it should state what the equivalence classes are and for use case testing it should contain the use cases being tested. This should not be a long document but should contain a minimum information about all the technical requirements.
- Your team's GitHub must remain public for the duration of ENSE 375, at a minimum. Remember, your GitHub is your portfolio of work for future employers to understand the type of experiences you have had, so keeping it public for long-term is not a terrible idea.
- All the code and test suits should be well commented.

- 6. Project Deliverables:** Besides fulfilling the presentation requirements, following will be delivered while following the given schedule
- a. A functional and systematically tested prototype.
 - b. Technical report (REPORT.md).
 - c. Testing report (TESTING.md).

S. No.	Deliverables	REPORT.md Components	Timeline	Date	Weight
1.	Problem Definition	Section 2.1 (in project file template)	3 rd Week	Sept. 17	10%
2.	Design Constraints and Requirements	Section 2.2	4 th Week	Sept. 24	10%
3.	Iterative Engineering Design Process (Solution 1 and 2).	Section 3.1 – 3.2	6 th Week	Oct. 08	10%
4.	Final Design (Solution 3), Implementation and Testing	Section 3.3	13 th Week	Dec. 03	60%
5.	Collaborative Teamwork and Communication Skills	Section 4 - 5			10%

- Students are required to fill Teamwork and Project Management sections (Project_File_Template.pdf on UR courses) of the report (REPORT.md) simultaneously with every report component and should be updated regularly.
- Weekly meetings will be held to monitor the progress of the work
- For every late submission, 10% penalty will be imposed per day.
- All the deadlines are hard deadlines. All the artifacts should be submitted before 23:59:59.

7. Submission Instructions:

- **Due: December 07, 2026, 23:59:59**
- Be sure to commit often. I will use GitHub to review your solution.
- Please email me on URCourses the GitHub project names and the hashes of the commits you are submitting to make sure we are looking at the same code.
- Remember, I will mark the project using the rubric available on the UR Courses.

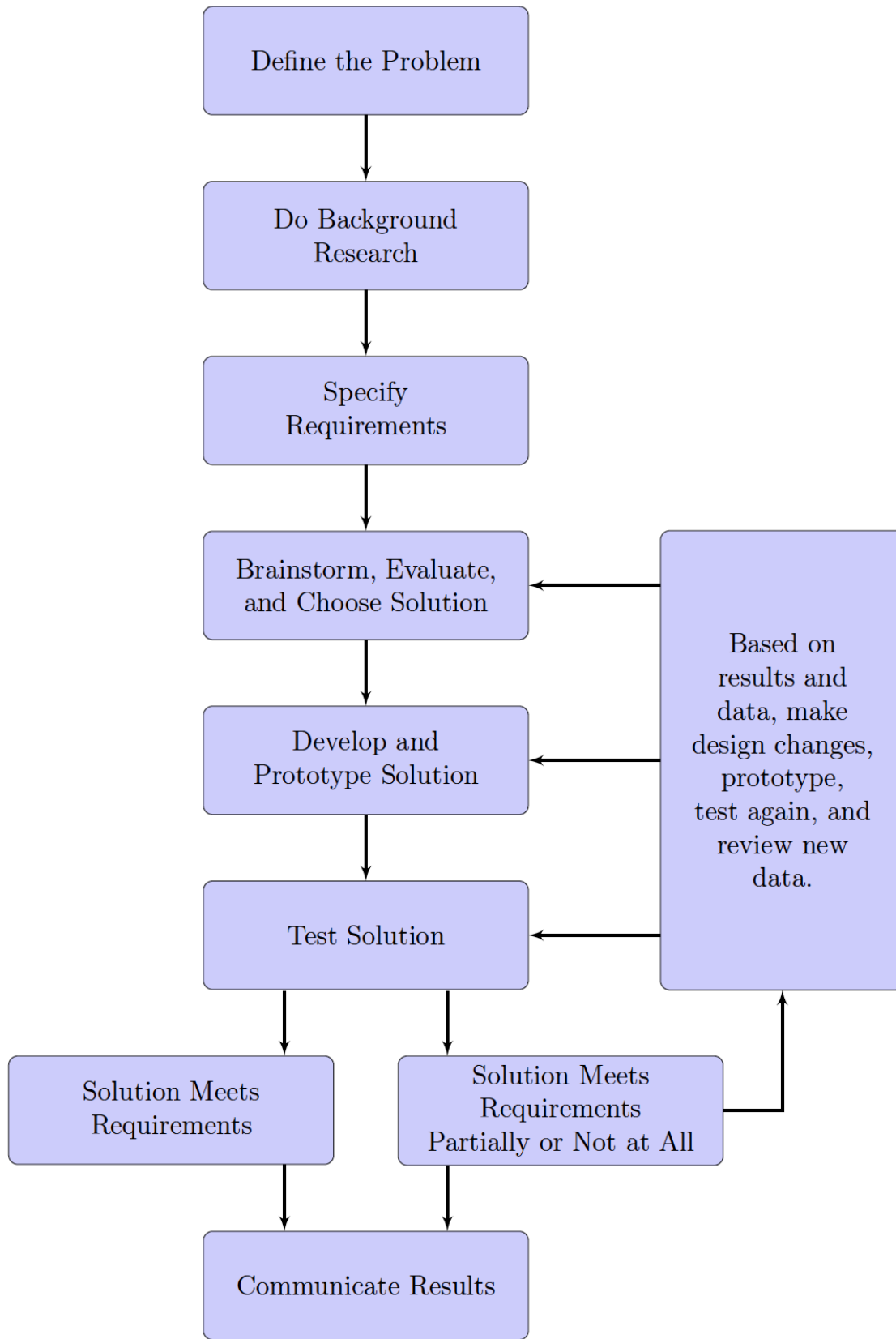


Figure 1: Engineering Design Process